SQL DATA ANALYSIS PROJECT

HR DATABASE MANAGEMENT SYSTEM

Comprehensive Analysis & Strategic Insights

Source: HR_DATABASE SQL Schema

December 2025

SYSTEM ARCHITECTURE

SYSTEM OVERVIEW

A centralized relational database management system designed to streamline workforce data, optimize organizational structure, and ensure data integrity across global operations.

7 Core Entities

The system is built on a normalized schema comprising seven primary tables that manage all aspects of HR data.

- regions
- countries
- locations
- jobs
- departments
- employees
- dependents

Strategic Purpose

Designed to serve as the single source of truth for all human resource related information within the organization.

- Centralize employee records
- Map organizational hierarchy
- Track global footprint
- Manage compensation data

System Scope

Comprehensive coverage of CRUD operations, reporting capabilities, and advanced analytics for decision making.

DATA INTEGRITY
Constraints & Triggers

ANALYTICS
Aggregations & Joins

SECURITY
Referential Rules

SCALABILITY
Normalized Design

HR DATABASE METRICS

KEY STATISTICS AT A GLANCE

High-level overview of the organizational data entities managed within the system.

40
TOTAL EMPLOYEES

11
DEPARTMENTS

19
JOB TITLES

30
DEPENDENTS TRACKED

$2k — $24k
MONTHLY SALARY RANGE

4
GLOBAL REGIONS

25
COUNTRIES

7
ACTIVE LOCATIONS

GLOBAL FOOTPRINT

GEOGRAPHIC DISTRIBUTION

Analysis of workforce distribution across 4 major regions and 7 strategic office locations.

Americas
Region 2

- Southlake, Texas (US)
- S. San Francisco, CA (US)
- Seattle, WA (US)
- Toronto, ON (CA)

Europe
Region 1

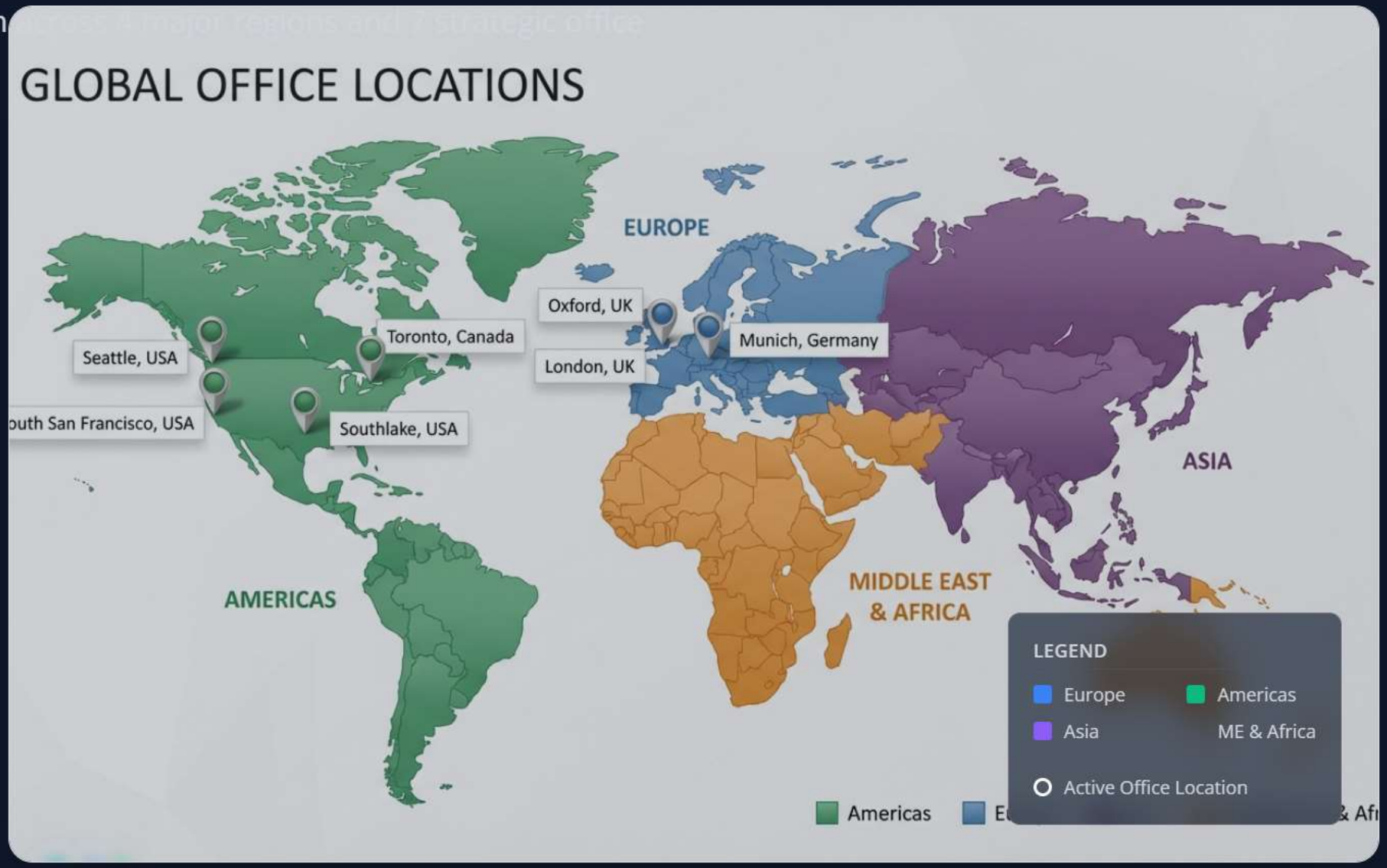
- London (UK)
- Oxford (UK)
- Munich (DE)

Asia
Region 3

High growth potential region with emerging markets identified.

ME & Africa
Region 4

Strategic expansion zone. Currently monitoring markets.



ENTITY-RELATIONSHIP DESIGN

DATABASE ARCHITECTURE

A highly normalized Relational Database Management System (RDBMS) schema ensuring data consistency and minimizing redundancy.

Schema Highlights

Normalized Structure

Designed to 3rd Normal Form (3NF) to prevent data anomalies.

Core Dependency Chain

Regions → Countries → Locations → Departments → Employees.

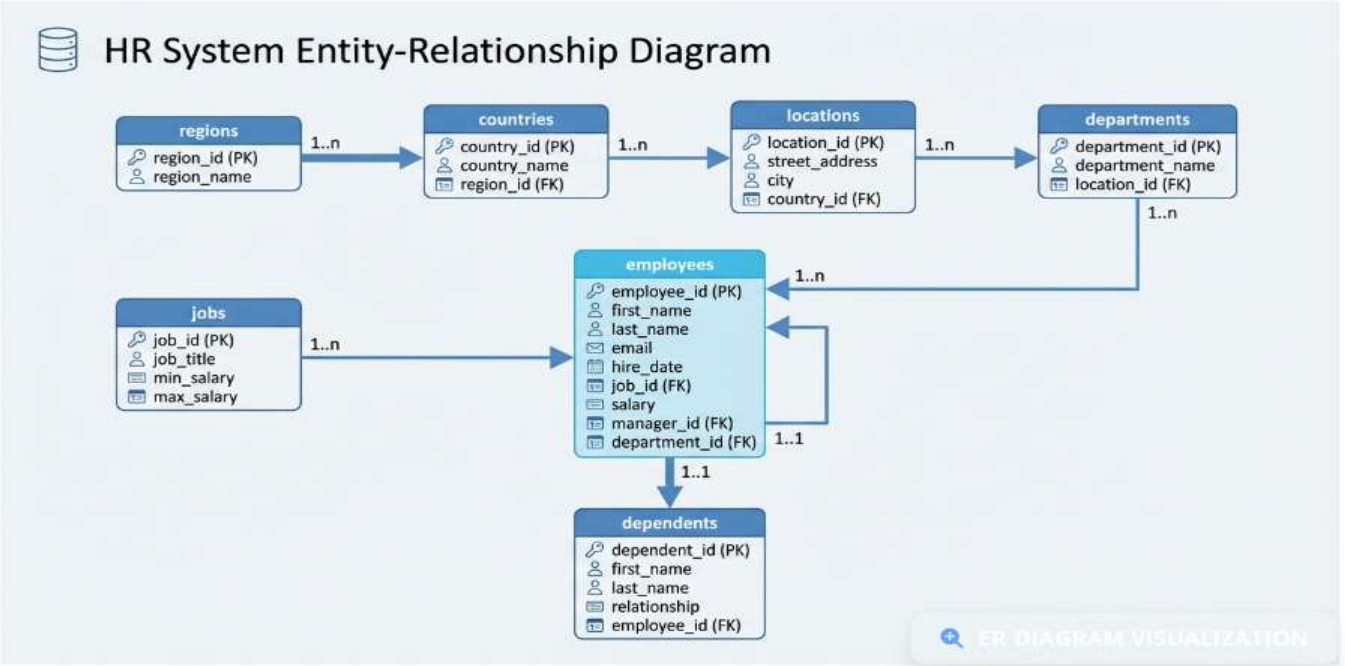
Self-Referencing

Employees table manages hierarchy via `manager_id`.

PRIMARY TABLES

- `regions (PK: region_id)`
- `countries (PK: country_id)`
- `locations (PK: location_id)`
- `departments (PK: dept_id)`
- `jobs (PK: job_id)`
- `employees (PK: emp_id)`
- `dependents (PK: dep_id)`

Primary Key (PK) Foreign Key (FK)



ER DIAGRAM VISUALIZATION

METHODOLOGIES & CONTROLS

DATA INTEGRITY

Implementing robust constraints, automated triggers, and referential safeguards to ensure database reliability.

Referential Integrity

Enforces strict relationships between parent and child tables, preventing orphaned records across the schema.

CASCADE Actions

Automated **ON DELETE** and **ON UPDATE** propagation ensures changes reflect instantly downstream.

Automated Triggers

Logic hooks (e.g., syncing dependent last names) maintain consistency without manual intervention.

Field Constraints

Mandatory fields (NOT NULL) and Default values preserve data quality at the entry point.

CASCADE LOGIC VISUALIZATION

Scenario: Updating a Region ID

When a primary key in the **Regions** table changes, the system automatically propagates this update through the entire hierarchy to maintain data validity.

`UPDATE regions SET id=10` — Action Initiated

`countries (region_id → 10)` — Auto-Updated

`Constraint Verified` — Integrity OK

```
FOREIGN KEY (region_id) REFERENCES regions (region_id)
  ON DELETE CASCADE
  ON UPDATE CASCADE
```

SQL METHODOLOGY

QUERY CATEGORIES

A comprehensive breakdown of the data retrieval strategies used to extract, organize, and analyze HR workforce information efficiently.

Retrieval & Sorting

The foundation of data analysis, allowing for precise column selection and organizing result sets based on criteria like hire date or salary.

SELECT	Extract specific columns
ORDER BY	Sort ASC / DESC
FROM	Target table source

Optimization & Limits

Techniques to remove redundancy and restrict output volume, essential for identifying top performers or unique department lists.

DISTINCT	Remove duplicates
TOP N	Limit result rows
WHERE	Conditional filtering

Pagination Logic

Sophisticated window functions enabling sequential data access, crucial for application interfaces and handling large employee datasets.

```
ROW_NUMBER() OVER (
  ORDER BY employee_id
) AS rn

// Enable row-based navigation
```

DATA SELECTION LOGIC

FILTERING TECHNIQUES

Precision data extraction using standard SQL comparison operators and the WHERE clause to isolate relevant records.

The WHERE Clause

Acts as a gatekeeper for your queries, evaluating conditions for each row. Only records that satisfy the condition (return `TRUE`) are included in the result set.

COMPARISON OPERATORS

=	Equal to	<>	Not Equal
>	Greater Than	<	Less Than
≥	Greater/Equal	≤	Less/Equal

exact_match_query.sql

```sql
-- Find specific employee by Last Name
SELECT * FROM employees
WHERE last_name = 'Chen';
```

salary_filter.sql

```sql
-- Filter high earners and sort descending
SELECT first_name, salary FROM employees
WHERE salary > 14000
ORDER BY salary DESC;
```

date_logic.sql

```sql
-- Employees hired after 1999 excluding Dept 8
SELECT * FROM employees
WHERE hire_date > '1999-01-01'
AND department_id <> 8;
```

ADVANCED QUERY LOGIC

ADVANCED OPERATORS

Expanding beyond basic filtering with logical combinations, pattern matching, and special handling for null values and duplicates.

OPERATOR	FUNCTIONALITY	SYNTAX LOGIC	HR EXAMPLE
AND OR	Combines multiple conditions. **AND** requires all true; **OR** requires at least one true.	`cond1 AND cond2` `cond1 OR cond2`	`WHERE dept_id = 5` `AND salary > 5000`
BETWEEN IN	Simplifies range checks and multiple OR conditions. Inclusive of endpoints.	`val BETWEEN x AND y` `col IN (a, b, c)`	`WHERE salary` `BETWEEN 9000 AND 12000`
LIKE	Matches string patterns using wildcards: `%` (0+ chars) and `_` (1 char).	`col LIKE 'pattern%'` `col LIKE '_a%'`	`WHERE last_name` `LIKE '%an%'`
IS NULL	Identifies missing or undefined data. Standard operators (=) cannot check for NULL.	`col IS NULL` `col IS NOT NULL`	`WHERE phone_number` `IS NULL`
DISTINCT	Removes duplicate rows from the result set, returning only unique combinations.	`SELECT DISTINCT col1`	`SELECT DISTINCT job_id` `FROM employees`

COMPENSATION METRICS

SALARY ANALYSIS

Visualizing compensation distribution across salary bands to identify pay structures and outliers.

Salary Bands

High Income > $5,000	Top Tier
Average Income $3,000 - $5,000	Mid Tier
Entry Level < $3,000	Base Tier

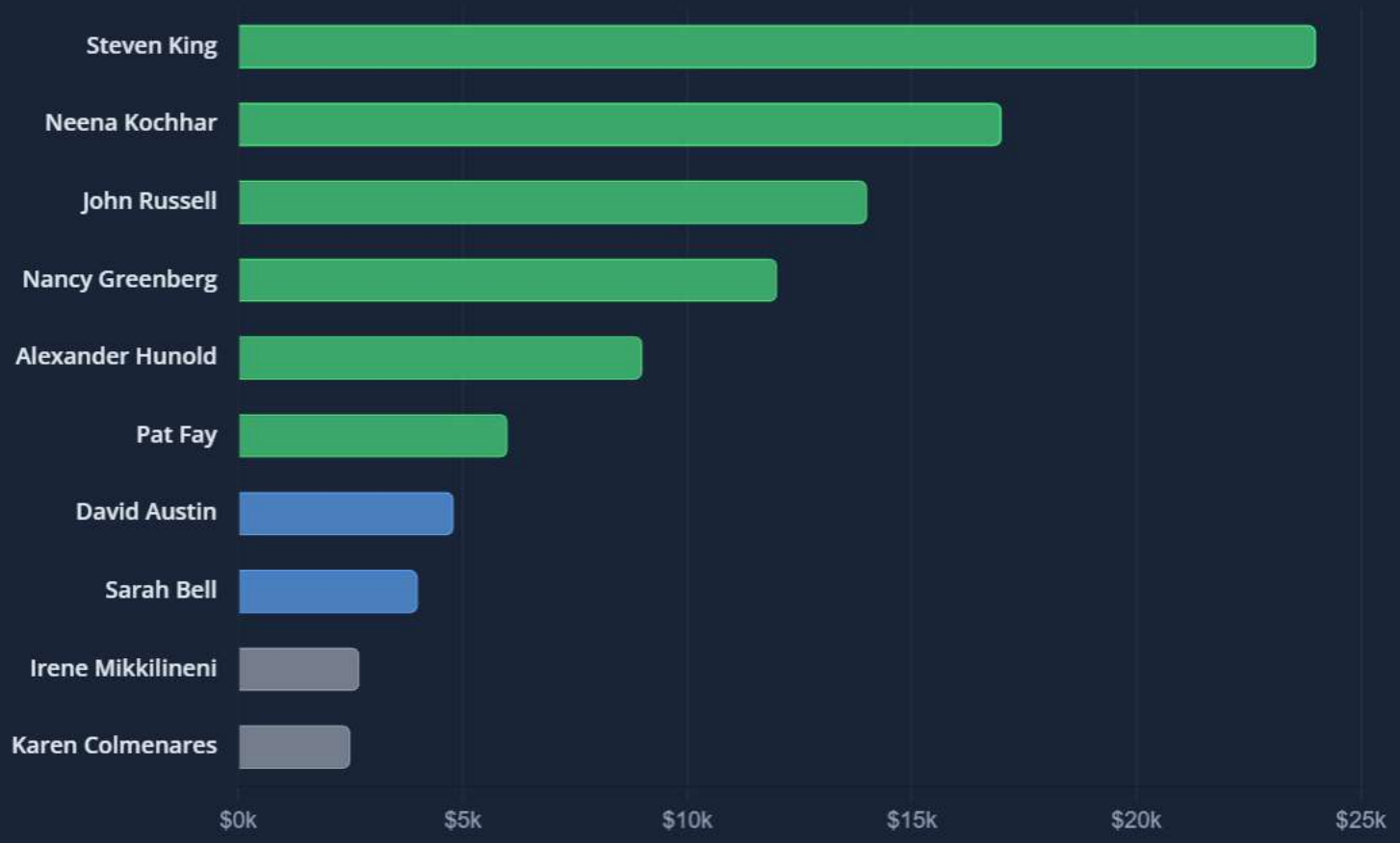
TOP EARNER

Steven King

Role	Salary
President	$24,000

Representative Salary Distribution

Selected Roles



WORKFORCE ANALYTICS

DEPARTMENT INSIGHTS

Comparative analysis of resource distribution and compensation structures across organizational units.

TOTAL DEPTS	TOTAL STAFF
11	40

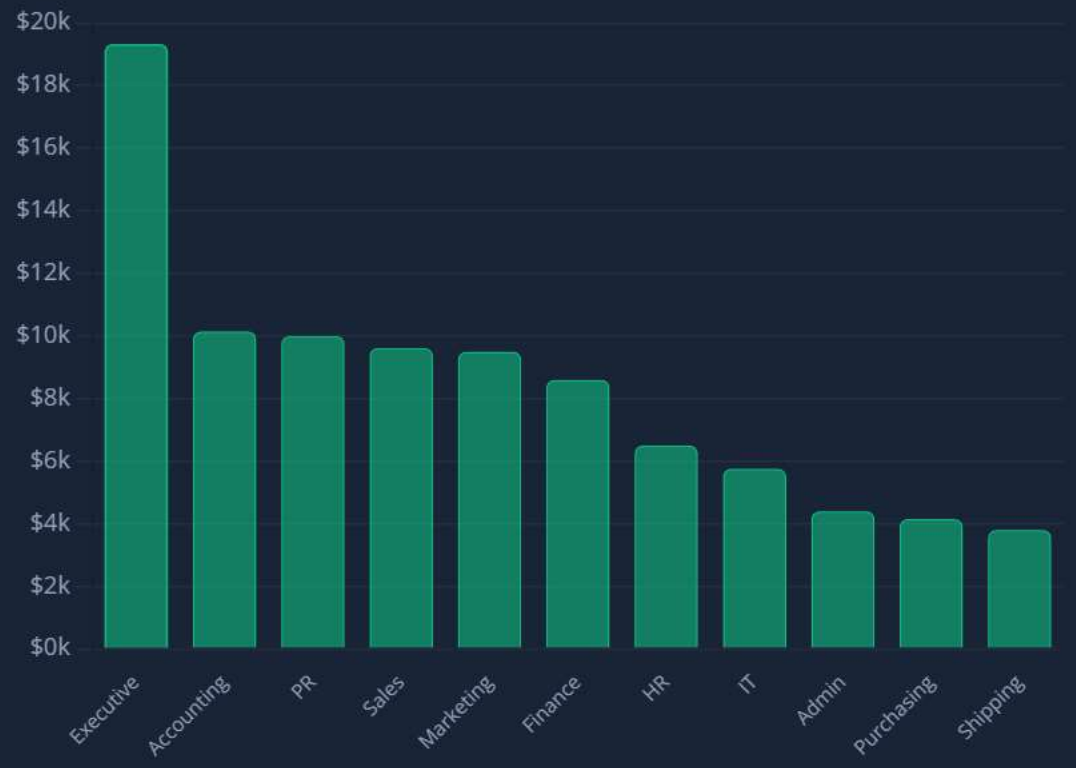
Headcount Distribution

By Department

Shipping and **Sales** departments comprise largest workforce segments.

Average Salary

USD / Month



Executive team leads compensation; **Purchasing** & **Shipping** maintain cost efficiency.

DATA INTEGRITY REPORT

DATA QUALITY SPOTLIGHT

Identification of critical data gaps, inconsistencies, and opportunities for enrichment within the current dataset.

DETECTED ANOMALIES

Connectivity Gaps

HIGH PRIORITY

Multiple employee records in the **Sales Department** contain `NULL` values in the `phone_number` field.

→ *Impact: Hinders direct communication with field representatives.*

Geographic Precision

MEDIUM PRIORITY

European office locations (UK, Germany) frequently miss `state_province` data. While valid locally, this creates inconsistency in global reporting.

→ *Opportunity: Standardize region mapping for non-US addresses.*

Data Hygiene

MAINTENANCE

Potential for whitespace characters or inconsistent casing in email addresses. Suggest implementing a lower-case + trim constraint trigger.

HEALTH & REMEDIATION

85%

QUALITY SCORE

Overall database health is **Robust**, with isolated completeness issues in the Sales schema.

Action Plan

- [x] Mandatory field constraint for Phone Number
- [ ] Backfill missing State/Province data
- [ ] Implement email validation trigger

GLOBAL STRATEGY

EXPANSION OPPORTUNITIES

Strategic gap analysis revealing 21 untapped markets across 4 global regions based on current HR database footprint.

25

Current Footprint

Active Hubs **4**

Growth Potential

Untapped Markets **21**

Strategic Insight: 84% of monitored countries have no physical office presence, with Asia and Africa completely devoid of operational hubs.

Europe

25% Coverage (2/8)

- Germany
- UK
- Belgium
- Switzerland
- Denmark
- France
- Italy
- Netherlands

Americas

40% Coverage (2/5)

- Canada
- USA
- Argentina
- Brazil
- Mexico

Asia

0% Coverage (0/6)

High Priority

- Australia
- China
- Hong Kong
- India
- Japan
- Singapore

Middle East & Africa

0% Coverage (0/6)

Future Market

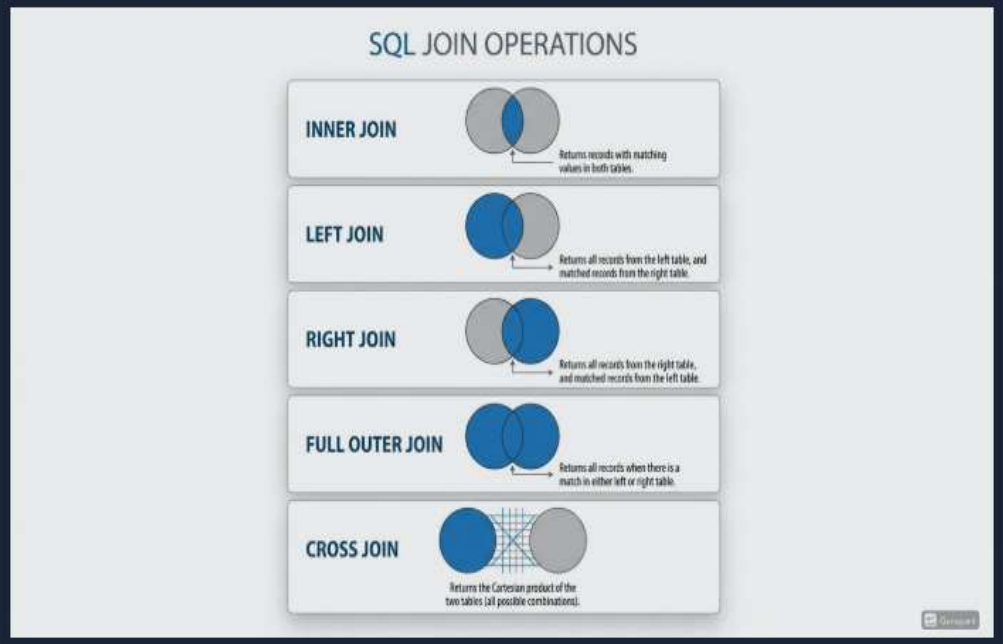
- Egypt
- Israel
- Kuwait
- Nigeria
- Zambia
- Zimbabwe

RELATIONAL LOGIC

JOIN OPERATIONS

Combining data rows from two or more tables based on related columns to create unified datasets.

Visual Guide



INNER JOIN

Matching Rows Only

Returns records that have matching values in both tables (Employees assigned to Depts).

```
SELECT e.name, d.dept_name
FROM employees e
JOIN departments d
ON e.dept_id = d.dept_id;
```

LEFT JOIN

All Left, Match Right

Returns all records from left table, and matches from right (Countries even without Offices).

```
SELECT c.name, l.city
FROM countries c
LEFT JOIN locations l
ON c.country_id = l.country_id;
```

RIGHT JOIN

All Right, Match Left

Opposite of LEFT JOIN. Rarely used; usually restructured as LEFT JOIN for clarity.

```
SELECT e.name, d.dept_name
FROM employees e
RIGHT JOIN departments d
ON e.dept_id = d.dept_id;
```

FULL OUTER

Match Either

Returns all records when there is a match in either left or right table record.

```
SELECT f.fruit, b.basket
FROM fruits f
FULL OUTER JOIN baskets b
ON f.basket_id = b.id;
```

STATISTICAL ANALYSIS

AGGREGATION & GROUPING

Synthesizing raw data into meaningful metrics using aggregate functions and conditional grouping logic.

CORE FUNCTIONS

COUNT(*)
Returns the number of rows matching criteria.

SUM(col)
Calculates total value of a numeric column.

AVG(col)
Computes the average value of a numeric column.

MIN() / MAX()
Returns the smallest or largest value in a set.

APPLIED SCENARIOS

Scenario: Headcount per Department

GROUP BY

Group employees by their department identifier to count staff levels.

```
SELECT department_id, COUNT(*) AS headcount
FROM employees
GROUP BY department_id
ORDER BY headcount DESC;
```

Scenario: High-Budget Departments

HAVING

Filter groups based on an aggregate condition (Total Salary > $20,000).

```
SELECT department_id, SUM(salary) AS total_salary
FROM employees
GROUP BY department_id
HAVING SUM(salary) > 20000;
```

ADVANCED SQL LOGIC

QUERY TECHNIQUES

Implementing complex business logic using subqueries, conditional expressions, and existence checks.

Conditional Logic (CASE)

Transforming data into categories dynamically.

```
SELECT first_name, salary, CASE WHEN salary < 3000 THEN 'Low' WHEN
salary BETWEEN 3000 AND 5000 THEN 'Avg' ELSE 'High' END AS salary_band
FROM employees;
```

Useful for creating histograms or groupings not present in the raw data.

Scalar Subqueries

Comparing rows against calculated aggregates.

```
-- Employees earning above company average
SELECT first_name, salary FROM employees WHERE salary > ( SELECT
AVG(salary) FROM employees );
```

The subquery executes once, returning a single value for comparison.

Existence Check (EXISTS)

Testing for relationships without returning extra data.

```
-- Find employees with dependents
SELECT * FROM employees e WHERE EXISTS ( SELECT 1 FROM dependents d
WHERE d.employee_id = e.employee_id );
```

Often more performant than IN or JOIN when just validating existence.

Set Comparison (ALL / ANY)

Comparing a value against a list of results.

```
-- Salary higher than everyone in Dept 8
SELECT first_name, salary FROM employees WHERE salary > ALL ( SELECT
salary FROM employees WHERE department_id = 8 );
```

Equivalent to finding value > MAX() of the subquery result set.

EXECUTIVE SUMMARY

KEY TAKEAWAYS

Comprehensive evaluation of the HR system's architecture, workforce distribution, and data integrity.

Robust Architecture

The 7-table normalized schema with strict referential integrity and `CASCADE` operations ensures reliable data management and supports complex querying needs.

Organizational Clarity

Effective hierarchical tracking of 40 employees across 11 functional departments. Self-referencing manager IDs maintain clear reporting lines from the President down.

Compensation Insights

Defined salary bands ($2k–$24k) allow for precise role benchmarking. Analysis reveals clear distribution patterns, aiding in future budget planning and equity adjustments.

Improvement Areas

While the core structure is sound, specific gaps in phone number records and address standardization (State/Province) present immediate opportunities for data enrichment.

FUTURE ROADMAP

STRATEGIC RECOMMENDATIONS

Prioritized action plan to optimize database integrity, expand operational reach, and modernize HR processes.

IMMEDIATE

Enhance Data Quality

Address critical gaps in contact information and standardize address formats to ensure communication reliability.

- Enforce NOT NULL constraints on Phone Numbers
- Standardize state/province codes globally
- Implement email format validation

SHORT TERM

Expand Footprint

Leverage existing monitoring in 21 non-office countries to establish new strategic regional hubs.

- Prioritize Asia (Singapore, Tokyo) entry
- Evaluate Americas expansion (Brazil)
- Convert remote presence to physical offices

QUARTERLY

Optimize Comp

Address department-level disparities and align lower-quartile earners with market standards.

- Review sub-$3k salaries in Shipping
- Standardize pay bands by job function
- Audit executive-to-staff pay ratios

CONTINUOUS

Automate Logic

Implement database triggers to maintain relational integrity and reduce manual administrative overhead.

- Auto-sync dependent last names (Trigger)
- Cascade delete for departed managers
- Automate salary band assignment